

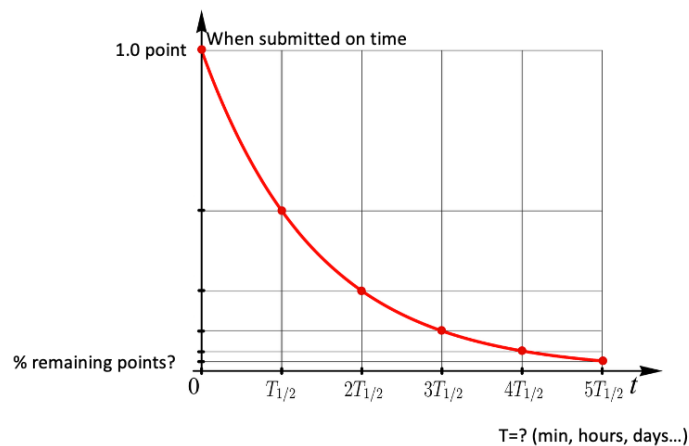
PHY-275 Fall 2026 Assignment #3 due 9/23/26 9:00pm

You will receive 1 credit upon complete all the tasks.

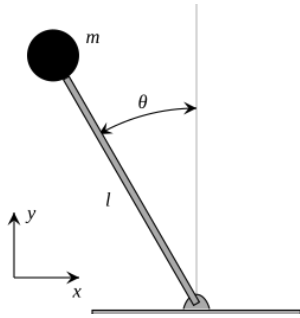
Task #1: Help me derive the “point degradation” curve for late submission:

Please refer to <https://en.wikipedia.org/wiki/Half-life> for more info.

Please derive the governing equation in the form of 1st order linear ODE, then find the general solution with the initial condition $T(0)=1$. Pick reasonable parameters so your model will be practical for this application.



Task #2: Practice your “Eigenskills”:

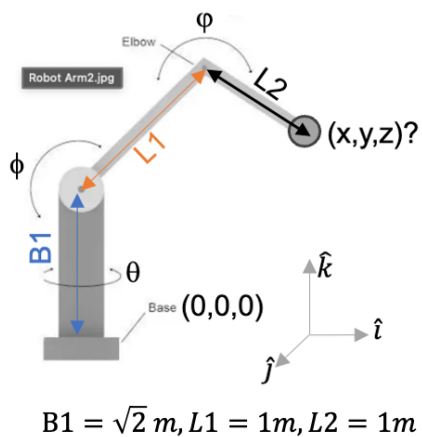


Inverted pendulum is a classic: The pendulum is assumed to consist of a point mass (m), affixed to the end of a massless rigid rod, of length ℓ , attached to a pivot point at the end opposite the point mass.

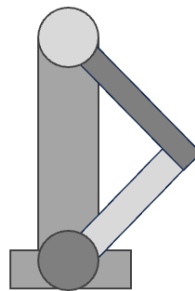
- 1) Derive the equation of motion of this system
- 2) Convert this 2nd order ODE to state space (system of equations in matrix form)
- 3) Assume $\ell = 9.8$ m, find its eigenvalue, eigenvector, and the phase diagram.

Task #3: Practice your Numpy skills, download the companion Colab notebook from the course site and work out all the problems there.

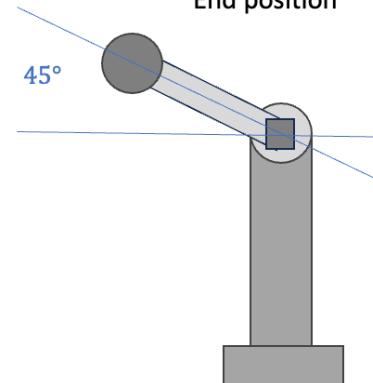
- 1) Check your eigen calculation in task #2
- 2) Check your phase diagram in task #2
- 3) Check your DH parameters of the robotic arm.



Starting position



End position



- 3.1) Figure out the end position analytically;
- 3.2 Then find out the turning angle of each arm in DH parameter form;
- 3.3) Validate your math in Numpy. Please remember to paste screen shot of all three Numpy tasks.